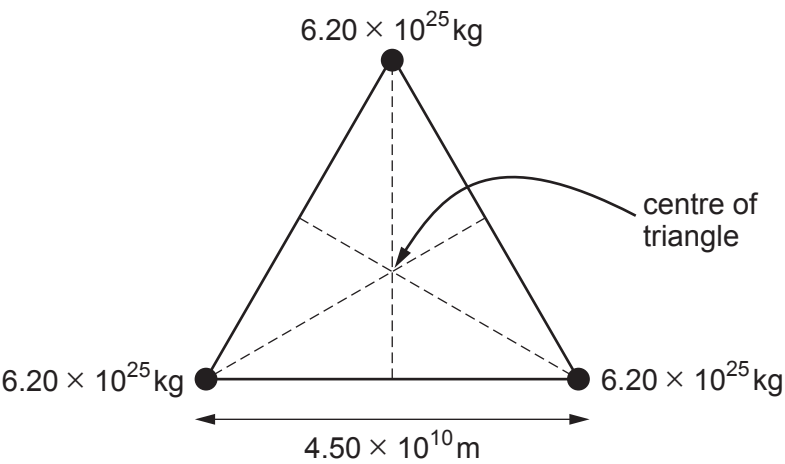


3. Three planets of equal mass are arranged in an **equilateral triangle** as shown.



(a) (i) Show that the gravitational force exerted by one planet on another is approximately  $1.3 \times 10^{20} \text{ N}$ . [2]

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(ii) Hence, explain why the resultant gravitational force experienced by any of the planets is approximately  $2 \times 10^{20} \text{ N}$  towards the centre of the triangle. [4]

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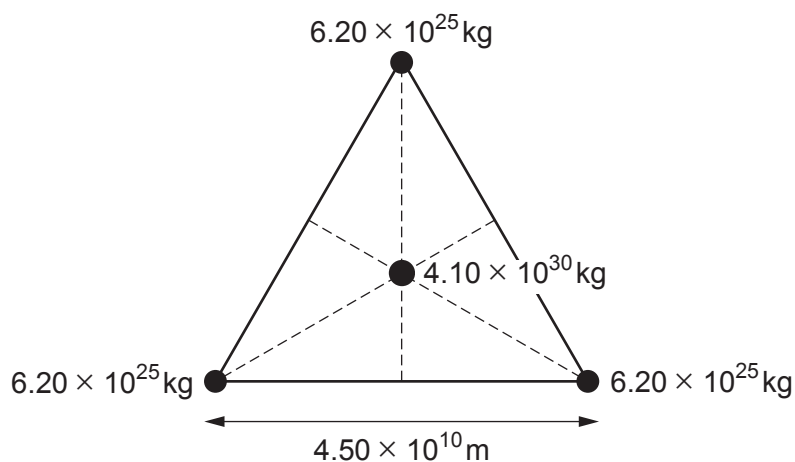
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- (b) A star is located at the centre of the triangle.



- (i) By adding to the diagram, show that the distance between the star and any of the planets is given by: [2]

$$\frac{4.5 \times 10^{10}}{2 \times \cos 30^\circ}$$

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- (ii) Rhodri states that the force calculated in part (a)(ii) is negligible compared with the gravitational force exerted on a planet by the star. Determine whether or not Rhodri is correct. [2]

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- (c) (i) Show that the gravitational potential at the position of a planet due to the star and the other two planets is approximately  $-1 \times 10^{10} \text{ J kg}^{-1}$ . [3]

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- (ii) Rhodri now states that, if an object with kinetic energy  $7 \times 10^{35} \text{ J}$  struck one of the planets, the planet would gain enough energy to escape the gravitational pull of the star. Discuss to what extent Rhodri's statement might be correct. [3]

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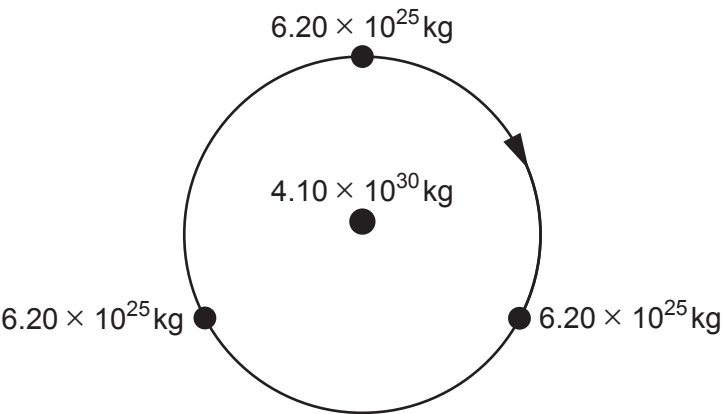
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- (d) The three planets, whilst orbiting the star, stay in an equilateral triangle. Delyth claims that these planets cannot be detected by the periodic Doppler shift of the star. Determine whether or not Delyth is correct. [3]



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